

**GIS-BASED OPTIMAL AMBULANCE ROUTING AND EMERGENCY
RESPONSE TIME ANALYSIS USING NETWORK ANALYSIS**

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CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

Emergency medical services (EMS) play a critical role in ensuring timely medical intervention, where even slight delays in response time can significantly affect patient survival outcomes. In urban environments characterized by rapid population growth, traffic congestion, and complex road networks, the efficiency of ambulance routing has become a major concern. Traditional dispatch systems often rely on simple distance-based routing or human judgment, which fails to account for dynamic road conditions, network constraints, and travel time variability. As a result, there is an increasing need for intelligent systems that can optimize ambulance routing using advanced spatial and computational techniques.

Recent advancements in Geographic Information Systems (GIS) and network analysis have provided powerful tools for modeling transportation networks and improving emergency response efficiency. Studies such as Mappouras et al. (2025) demonstrate the integration of GIS with Artificial Intelligence to create decision-support systems capable of optimizing ambulance dispatch and routing using real-time data. Similarly, Alamri (2023) developed a spatial routing system that combines Voronoi-based accessibility modeling with time-dependent Dijkstra algorithms to improve emergency service delivery. These approaches emphasize that response time, rather than distance, is the most critical factor in emergency routing.

Further research has highlighted the importance of network-based analysis in understanding accessibility and optimizing routes. Paul-Silviu NICOARĂ (2014) demonstrated how GIS-based shortest path algorithms can effectively determine

optimal routes to emergency facilities while accounting for road restrictions and infrastructure limitations. Other studies, such as Hao *et al.* (2024), have shown that spatial variables like road topology and urban layout significantly influence emergency response performance. Additionally, predictive and optimization-based approaches, including Ant Colony Optimization and machine learning models, have been explored to improve routing efficiency under dynamic conditions (Luo *et al.*, 2026; Métwalli *et al.*, 2026).

Beyond routing, spatial analysis techniques have also been applied to understand the distribution and dynamics of emergency incidents. For instance, Pérez-Fernández *et al.* (2025) and Aporto *et al.* (2026) utilized spatio-temporal mapping and hotspot analysis to identify patterns in emergency calls, providing insights for resource allocation and planning. Similarly, Payares-Garcia *et al.* (2023) developed stochastic models to predict emergency demand patterns, while Luo *et al.* (2026) introduced advanced accessibility models to evaluate healthcare service coverage in urban areas. These studies highlight the importance of integrating spatial analysis with decision-making frameworks for improved emergency management.

Despite these advancements, most existing systems are either highly complex, data-intensive, or reliant on real-time infrastructure that may not be readily available in developing regions. Many studies focus on advanced AI-driven or stochastic models, which, while effective, may not be easily implementable in practical academic or local government settings. Moreover, there remains a gap in developing simplified, GIS-based routing systems that can provide accurate, time-based routing and service coverage analysis without requiring extensive real-time data integration.

This study addresses this gap by proposing a GIS-based optimal ambulance routing system using network analysis techniques. The system focuses on travel-time-based routing, closest facility analysis, and service area modeling to evaluate emergency response efficiency. By leveraging accessible spatial data and established GIS tools, this research aims to provide a practical and scalable solution for improving ambulance dispatch and response time analysis.

The research problem addressed in this study is the inefficiency of current ambulance routing systems due to the lack of network-based, time-sensitive analysis. The goal of the study is to design and implement a GIS-based system that optimizes ambulance routing and evaluates emergency response coverage. It is hypothesized that the use of network analysis techniques based on travel-time impedance will significantly improve response time and service coverage compared to traditional distance-based methods.

The study is structured to include an introduction, a review of related literature, a detailed methodology outlining data preparation and network analysis techniques, followed by results, discussion, and conclusions. The methodology involves the use of GIS tools such as ArcGIS Network Analyst or QGIS plugins to construct a network dataset, perform routing analysis, and generate service area maps.

1.2 Statement of the Problem

Emergency response systems are highly dependent on efficient routing and timely dispatch of ambulance services. However, in many urban areas, ambulance routing decisions are still based on simple proximity measures or manual judgment, which do not reflect actual travel conditions. This results in increased response time, inefficient resource utilization, and unequal access to emergency services.

Previous studies have attempted to address this problem using various approaches. For instance, Alamri (2023) applied time-based routing algorithms to improve EMS accessibility, while Mappouras *et al.* (2025) integrated AI and GIS to optimize dispatch decisions. Similarly, Paul-Silviu NICOARĂ (2014) demonstrated the effectiveness of GIS-based network analysis in identifying optimal routes, and Rifka Alkhilyatul Ma'rifat (2024) highlighted the role of spatial factors in influencing response times. Advanced optimization techniques such as Ant Colony Optimization and predictive analytics have also been explored to enhance routing performance (Métwalli *et al.*, 2026; Muhammed & Okunola, 2020).

Despite these contributions, many existing studies rely on complex models, real-time data integration, or computationally intensive algorithms that may not be easily applicable in all contexts. There is still a lack of simplified, practical GIS-based systems that can effectively model ambulance routing using readily available spatial data. Additionally, limited attention has been given to integrating routing optimization with service area analysis to evaluate coverage and identify underserved regions.

This study aims to address these limitations by developing a GIS-based optimal ambulance routing system using network analysis techniques. The system will focus on travel-time-based routing, closest facility analysis, and service area modeling to provide a comprehensive framework for improving emergency response efficiency.

1.3 Aim and Objectives of the Study

The aim of this study is to design and implement a GIS-based optimal ambulance routing and emergency response time analysis system using network analysis techniques.

The specific objectives are to:

- i. develop a routable road network database with appropriate attributes such as travel time, road type, and traffic weighting.
- ii. determine optimal ambulance routes using shortest travel-time path algorithms within a GIS environment.
- iii. identify the closest ambulance facility to emergency incidents using network-based analysis.
- iv. analyze emergency response coverage using time-based service area modeling (5, 10, and 15 minutes).
- v. evaluate the performance of the proposed system by comparing it with conventional distance-based routing methods.

1.4 Methodology

This study adopts a GIS-based network analysis approach to optimize ambulance routing and analyze emergency response time. Spatial data, including road networks, ambulance station locations, and incident points, will be collected and processed within a GIS environment. The road network will be cleaned and structured into a network dataset with travel time defined as the impedance parameter. Network analysis techniques such as route analysis, closest facility analysis, and service area analysis will be applied to determine optimal routes and evaluate service coverage. The implementation will be carried out using ArcGIS Network Analyst or QGIS with appropriate plugins.

1.5 Significance of the Study

This study contributes to improving emergency medical service delivery by providing a practical and data-driven approach to ambulance routing optimization. The findings will assist emergency service providers in reducing response time, improving resource

allocation, and enhancing service coverage. It also contributes academically to the application of GIS in solving real-world urban and public health challenges.

1.6 Scope of the Study

The study focuses on ambulance routing within a defined urban area using road network data. The analysis is limited to static traffic conditions and does not incorporate real-time traffic data. The system evaluates routing efficiency and service coverage based on travel-time impedance.

1.7 Motivation for the Study

The increasing demand for efficient emergency response systems and the critical importance of reducing ambulance response time serve as the primary motivation for this study. The need for practical and implementable GIS-based solutions in developing regions further drives this research.

1.8 Justification of the Study

Given the limitations of traditional routing methods and the complexity of advanced optimization systems, there is a need for a balanced approach that is both effective and implementable. This study justifies the use of GIS-based network analysis as a reliable and accessible solution for improving ambulance routing and emergency response.

1.9 Organization of the Project

This project is organized into five chapters. Chapter One presents the introduction, including background, problem statement, and objectives. Chapter Two reviews relevant literature. Chapter Three describes the methodology and system design. Chapter Four presents results and analysis, while Chapter Five concludes the study and provides recommendations.

1.10 Definition of Terms

1. **GIS (Geographic Information System):** A system for capturing, storing, analyzing, and visualizing spatial data.
2. **Network Analysis:** A GIS technique used to model and analyze movement through a network such as roads.
3. **Impedance:** The cost associated with traveling through a network, typically measured in time or distance.
4. **Service Area:** A region reachable within a specified travel time from a facility.
5. **Closest Facility Analysis:** A method used to identify the nearest service location based on network travel cost.
6. **Emergency Response Time:** The time taken for an ambulance to reach an incident location after dispatch.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter presents a comprehensive review of existing literature relevant to ambulance routing optimization and emergency response systems. The review focuses on key thematic areas including Geographic Information Systems (GIS), network analysis, ambulance routing optimization techniques, emergency response time analysis, and spatial accessibility modeling. It synthesizes findings from previous studies to identify trends, limitations, and research gaps, which form the foundation for the proposed study.

2.2 Geographic Information Systems (GIS) in Emergency Services

Geographic Information Systems (GIS) have emerged as a critical tool in the management and optimization of emergency services due to their ability to integrate spatial and non-spatial data for decision-making. Several studies have demonstrated the effectiveness of GIS in modeling emergency response systems and improving operational efficiency.

Mappouras *et al.* (2025) developed a GIS-integrated decision support system that leverages spatial data and artificial intelligence to optimize ambulance dispatch. Their study highlights how GIS enables real-time visualization and improved coordination in emergency situations. Similarly, Popescu & Bărbulescu (2025) utilized GIS-based spatial analysis to determine optimal routes to emergency healthcare facilities, emphasizing the importance of spatial accuracy in route planning.

In addition, Pérez-Fernández *et al.* (2025) applied GIS techniques for spatio-temporal analysis of emergency incidents, revealing patterns that can inform strategic planning.

Aporto *et al.* (2026) further demonstrated the use of GIS in hotspot detection, enabling authorities to identify high-demand areas for emergency services.

Despite these advancements, many GIS-based systems are either too complex or rely heavily on real-time data infrastructure, limiting their applicability in resource-constrained environments. This highlights the need for simplified GIS-based solutions that can still provide effective routing and analysis.

2.3 Network Analysis in Transportation Systems

Network analysis is a fundamental component of GIS used to model transportation systems and optimize routing decisions. It enables the representation of road networks as interconnected nodes and edges, facilitating the computation of optimal paths based on defined impedance factors such as distance or time.

Alamri (2023) applied a time-dependent Dijkstra algorithm within a GIS framework to optimize emergency vehicle routing. The study demonstrated that travel-time-based routing significantly outperforms distance-based approaches in real-world scenarios. Similarly, Amoakoh *et al.* (2024) emphasized the role of road network topology and traffic conditions in influencing route efficiency.

Advanced optimization techniques such as Ant Colony Optimization have also been explored for routing problems. Métwalli *et al.* (2026) showed that bio-inspired algorithms can dynamically adapt to changing conditions, improving route selection. However, these methods often require high computational resources and complex implementation.

Overall, while network analysis provides a strong foundation for routing optimization, there remains a need for practical implementations that balance efficiency and simplicity.

2.4 Ambulance Routing Optimization Techniques

Ambulance routing optimization has been widely studied using various computational and analytical techniques. Traditional approaches often rely on shortest path algorithms such as Dijkstra and A*, while modern approaches incorporate machine learning and heuristic optimization methods.

Luo et al. (2026) explored machine learning-based routing systems that predict optimal routes based on historical traffic data. Similarly, Muhammed & Okunola (2020) applied predictive analytics to improve emergency response strategies. These approaches demonstrate the potential of intelligent systems in enhancing routing efficiency.

However, many of these methods depend on large datasets and real-time data integration, which may not always be available. In contrast, GIS-based network analysis provides a more accessible alternative that can still achieve significant improvements in routing performance.

2.5 Emergency Response Time Analysis

Emergency response time is a critical performance metric in healthcare systems, directly impacting patient survival rates. Several studies have focused on analyzing and reducing response times using spatial and computational methods.

Ragsdale & Russell (2022) developed an accessibility-based model to evaluate emergency response coverage, highlighting the importance of travel time as a key determinant. Prasetyo Wibowo Yunanto (2025) introduced stochastic modeling techniques to predict response times under uncertainty.

These studies emphasize that response time analysis must account for multiple factors, including road conditions, traffic patterns, and spatial distribution of facilities. However, many models are complex and difficult to implement in practice, reinforcing the need for simpler, GIS-based approaches.

2.6 Spatial Accessibility and Service Area Analysis

Spatial accessibility analysis evaluates the ease with which emergency services can reach different locations within a given area. Service area modeling is commonly used to determine coverage zones based on travel time thresholds.

Alamri (2023) utilized Voronoi diagrams to model accessibility regions, while Urfali & Eymen (2025) applied advanced accessibility metrics to assess healthcare coverage. These approaches provide valuable insights into service distribution and help identify underserved areas.

Service area analysis within GIS allows for the visualization of coverage zones, typically represented as travel-time buffers (e.g., 5, 10, 15 minutes). This method is particularly useful for planning and resource allocation.

2.7 Conceptual Framework

The conceptual framework of this study illustrates the relationship between spatial data inputs, network analysis processes, and emergency response outputs.

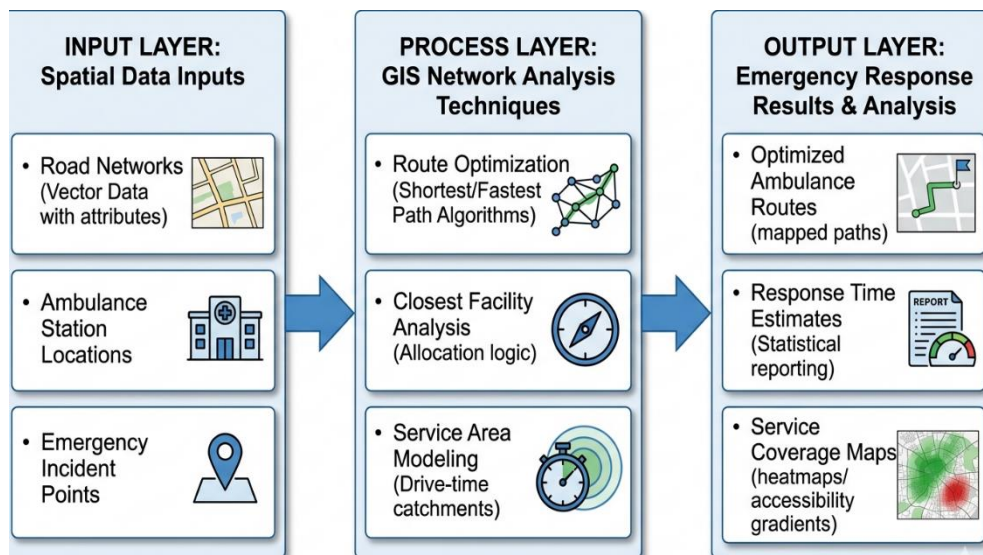


Figure 2.1: Conceptual Framework of the Study

The conceptual framework consists of three main components: input, process, and output. The input layer includes spatial datasets such as road networks, ambulance

station locations, and emergency incident points. The process layer involves network analysis techniques including route optimization, closest facility analysis, and service area modeling. The output layer consists of optimized routes, response time estimates, and coverage maps.

Figure 2.1 illustrates how spatial data is transformed through GIS-based network analysis into actionable insights for emergency response optimization. It highlights the flow from data acquisition to decision-making.

2.8 Theoretical Framework

This study is grounded in several key theoretical concepts:

1. **Graph Theory:** Road networks are modeled as graphs consisting of nodes (intersections) and edges (roads), enabling the application of shortest path algorithms.
2. **Dijkstra's Algorithm:** A fundamental algorithm used to compute the shortest path between nodes in a network based on defined cost functions.
3. **Location Theory:** This theory focuses on optimal placement of facilities to minimize travel cost and maximize accessibility.
4. **Accessibility Theory:** Emphasizes the importance of travel time in determining the effectiveness of service delivery.

These theories collectively provide the foundation for GIS-based network analysis and routing optimization.

2.9 Related Works

This section provides an overview of existing research relevant to the development of ambulance routing optimization and emergency response systems using Geographic Information Systems (GIS) and network analysis techniques. The reviewed literature captures the methodologies adopted by various researchers, including spatial analysis,

shortest path algorithms, accessibility modeling, and optimization techniques. It also highlights the datasets, analytical tools, and key findings presented in these studies. Furthermore, the review identifies notable limitations in existing approaches, which motivate the development of more practical, efficient, and implementable routing systems.

The literature used in this study was obtained from reputable academic databases such as IEEE Xplore, SpringerLink, Elsevier ScienceDirect, and ACM Digital Library. These sources include peer-reviewed journal articles, conference proceedings, and technical reports focusing on GIS applications, transportation network analysis, emergency response systems, and spatial optimization techniques. The use of these credible sources ensures the reliability and academic strength of the reviewed works.

Table 2.1: Summary of Related Works

Author & Year	Title	Methodology	Strength	Weakness
(Mappouras et al., 2025)	GIS-based Optimization	EMS GIS + AI	High accuracy	Complex implementation
(Alamri, 2023)	Time-based EMS Routing	Dijkstra + GIS	Realistic routing	Requires detailed data
(Paul-Silviu 2014)	NICOARĂ, GIS Routing Analysis	Network Analysis	Simple and effective	Limited dynamic factors
(Hao et al., 2024)	Spatial EMS Analysis	Spatial Modeling	Considers urban factors	Computational complexity
(Métwalli et al., 2026)	Ant Colony Optimization	Heuristic Algorithm	Adaptive routing	High computation cost
(Luo et al., 2026)	ML-based Routing	Machine Learning	Predictive capability	Data intensive

(Pérez-Fernández et al., 2025)	Spatio-temporal EMS	GIS Analysis	Pattern detection	Limited routing focus
(Aporto et al., 2026)	Hotspot Analysis	Spatial Statistics	Identifies demand zones	No routing optimization
(Wu & Yang, 2026)	Accessibility Modeling	GIS + Metrics	Coverage evaluation	Complex models
(Payares-Garcia et al., 2023)	Stochastic EMS Model	Probability Models	Handles uncertainty	Difficult implementation

2.10 Comments on Related Works

The reviewed literature demonstrates significant progress in the application of GIS, network analysis, and optimization techniques in emergency response systems. While advanced approaches such as machine learning and heuristic algorithms offer high accuracy and adaptability, they often require complex implementation and extensive data resources.

Conversely, GIS-based network analysis methods provide a more practical and accessible solution, particularly in environments with limited resources. However, many existing studies either focus solely on routing or accessibility without integrating both aspects into a unified framework.

This study addresses these limitations by combining route optimization, closest facility analysis, and service area modeling within a single GIS-based system. The proposed approach aims to balance simplicity, efficiency, and practical applicability, thereby contributing to both academic research and real-world emergency management.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter presents a comprehensive and detailed description of the methodology adopted for the development of the GIS-based optimal ambulance routing and emergency response analysis system. The study integrates spatial data modeling, graph theory, and network optimization techniques to design a system capable of minimizing ambulance response time. The methodology encompasses data acquisition, preprocessing, network dataset construction, algorithmic route optimization, closest facility determination, and service area analysis. In addition, mathematical formulations and algorithmic principles underlying the routing process are presented to provide a rigorous scientific foundation for the system.

3.2 Research Design

This study adopts a quantitative, spatial analytical, and system development research design, which combines Geographic Information Systems (GIS) with graph-based optimization techniques. The research is structured around transforming spatial data into a weighted graph representation, where intersections are modeled as nodes and road segments as edges.

Let the road network be represented as a graph:

$$G = (V, E)$$

where: V represents the set of nodes (intersections), and E represents the set of edges (road segments).

Each edge is assigned a weight corresponding to travel cost, defined primarily in terms of travel time:

$$w_{ij} = \frac{d_{ij}}{v_{ij}}$$

where: d_{ij} is the distance between node i and node j , and v_{ij} is the average travel speed along that segment.

This formulation allows the routing problem to be treated as a shortest path problem in a weighted graph, ensuring that the selected route minimizes travel time rather than distance.

3.3 Data Requirements and Preprocessing

The system requires high-quality spatial datasets, including road networks, ambulance station coordinates, and emergency incident locations. The preprocessing stage is critical to ensure the integrity and usability of the data.

Data preprocessing involves cleaning and structuring the spatial data to ensure topological correctness. This includes removing duplicate edges, correcting disconnected nodes, and ensuring that all road segments are properly connected. Additionally, attributes such as road length, speed limits, and road hierarchy are assigned.

Travel time for each road segment is computed using:

$$T_{ij} = \frac{L_{ij}}{S_{ij}}$$

where:

T_{ij} is the travel time

L_{ij} is road length, and

S_{ij} is speed.

This ensures that the impedance used in routing reflects real-world travel conditions.

3.4 Network Dataset Construction

The network dataset is constructed by converting the road network into a graph structure suitable for computational analysis. This involves defining connectivity rules, assigning impedance values, and enabling turn restrictions where applicable.

The adjacency matrix representation of the network is given as:

$$A = [a_{ij}]$$

Where:

$$a_{ij} = \begin{cases} w_{ij}, & \text{if there is a direct connection} \\ \infty, & \text{otherwise} \end{cases}$$

This matrix representation facilitates efficient computation of shortest paths using graph algorithms.

3.5 Route Optimization Model

The core of the system is the route optimization model, which determines the shortest travel-time path between an ambulance station and an emergency location. This is achieved using Dijkstra's Algorithm, a widely used method for solving shortest path problems in weighted graphs.

The objective function is defined as:

$$\min \sum_{(i,j) \in E} w_{ij} x_{ij}$$

Subject to:

$$\sum_j x_{ij} - \sum_j x_{ji} = \begin{cases} 1, & \text{if } i = s \\ -1, & \text{if } i = t \\ 0, & \text{otherwise} \end{cases}$$

Where:

x_{ij} indicates whether an edges is selected,

s is the source node, and

t is the destination node.

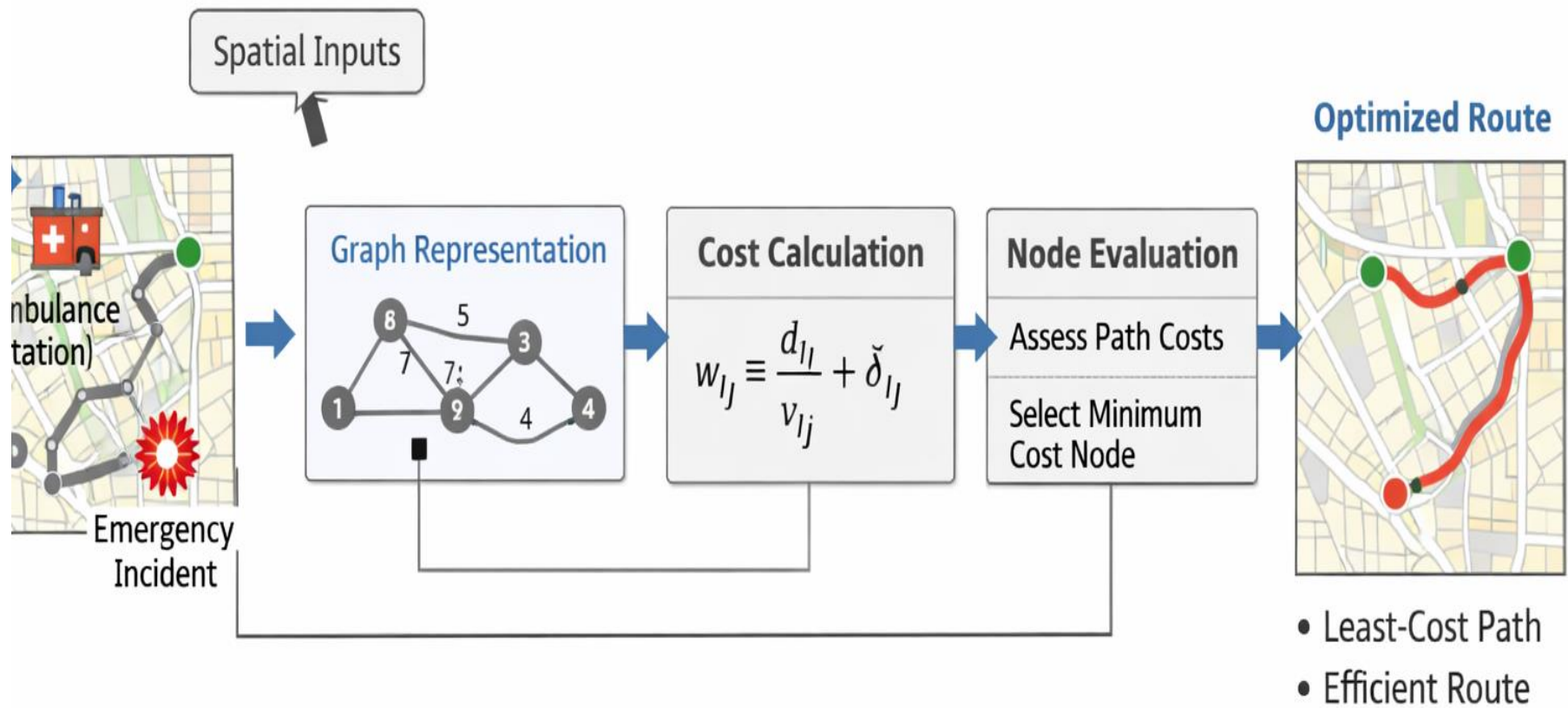


Figure 3.2: Route Optimization Process

Figure 3.2 illustrates the systematic process involved in determining the optimal route for ambulance dispatch using network analysis techniques. The process begins with the acquisition of spatial inputs, specifically the ambulance station (origin) and the emergency incident location (destination), which are mapped onto a geographic road network. These inputs are then transformed into a graph representation, where road intersections are modeled as nodes and road segments as edges, each associated with a travel cost such as time or distance.

Subsequently, cost calculation is performed using a weighted function that incorporates distance, speed, and possible delay factors, ensuring that the model reflects real-world traffic conditions. The node evaluation stage follows, where algorithms such as Dijkstra's algorithm iteratively assess neighboring nodes and update cumulative costs to identify the least-cost path. Finally, the optimized route is generated, representing the most efficient path between the origin and destination. This process ensures that ambulance movement is both time-efficient and reliable, thereby significantly improving emergency response performance.

3.6 Closest Facility Analysis Model

The closest facility analysis extends the routing model by considering multiple potential origins (ambulance stations) and identifying the most optimal one for responding to a given emergency. This is particularly important in real-world scenarios where multiple ambulances are distributed across different locations.

Mathematically, let $S = \{s_1, s_2, \dots, s_n\}$ represent the set of ambulance stations and i represent the incident location. The objective is to determine the station s_k that minimizes travel time:

$$s^* = \arg \min_{s_k \in S} d(s_k, i)$$

where $d(s_k, i)$ is the shortest path cost between station s_k and incident i , computed using the routing model described in Section 3.5.

This problem can be efficiently solved using a multi-source shortest path approach, where distances from all stations to the incident are computed simultaneously. Alternatively, the problem can be reversed by treating the incident as the source and computing distances to all stations.

In GIS implementation, this is handled using the “Closest Facility” solver, which evaluates all candidate facilities and returns the one with the minimum impedance value. This ensures that emergency response decisions are based on actual travel time rather than proximity alone.

Figure 3.3 illustrates the process of identifying the nearest ambulance facility to an emergency incident using network-based spatial analysis. The diagram begins with multiple ambulance stations (facilities) connected to a single incident location through different routes, each associated with a specific travel time. These travel times are computed using the underlying road network, taking into account factors such as distance, speed, and possible delays.

The model then performs a comparative evaluation of all possible routes from the facilities to the incident location. By analyzing the computed travel times, the system identifies the facility with the minimum response time. This selected facility represents the optimal choice for emergency dispatch. The process ensures that decision-making is based on actual accessibility rather than simple geographic proximity, thereby improving response efficiency. Overall, the model enhances emergency service delivery by ensuring that the fastest available ambulance is deployed to the incident location.

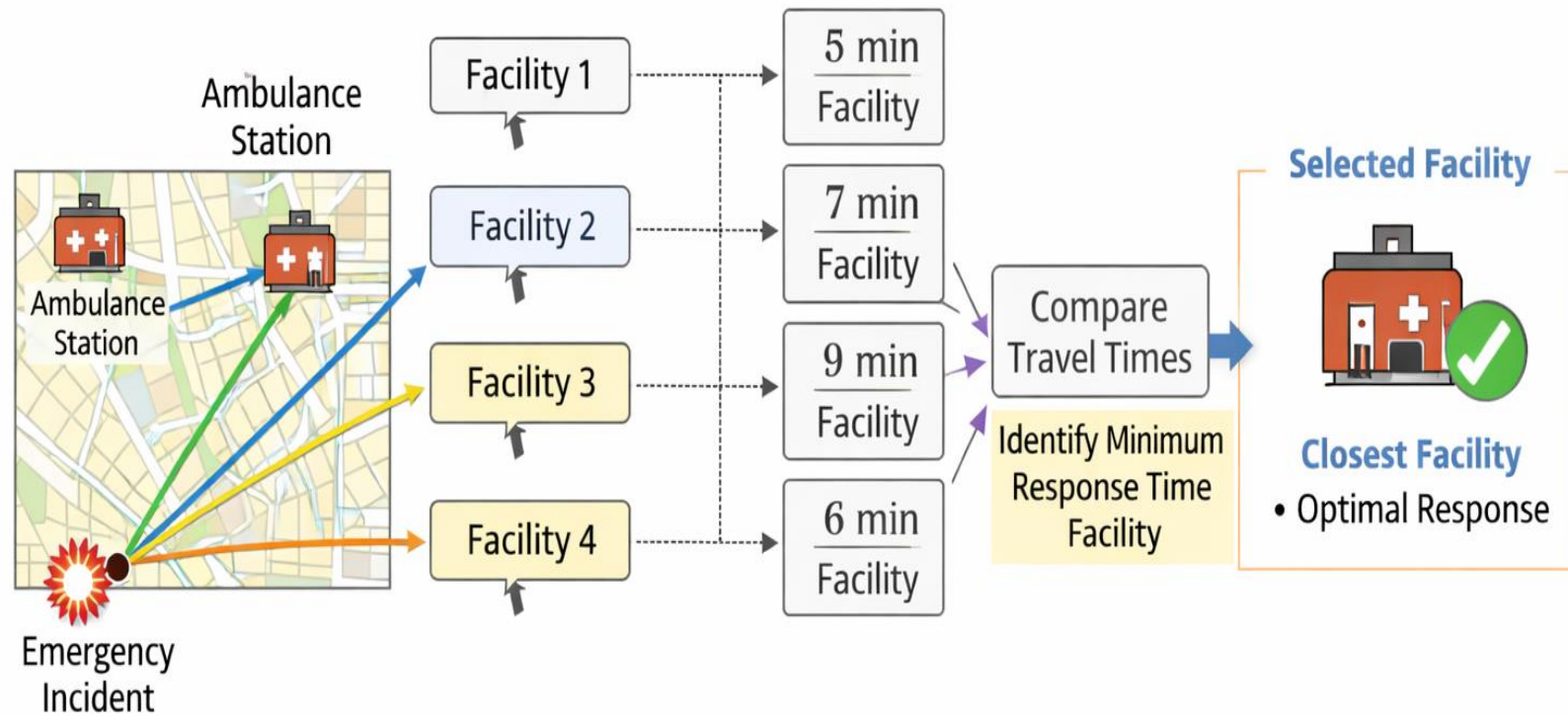


Figure 3.3: Closest Facility Analysis Model

3.7 Service Area Analysis Model

Service area analysis is used to evaluate the spatial coverage of ambulance services and determine the accessibility of emergency care within the study area. This analysis is based on the concept of reachable nodes within a given travel time threshold.

Let $s \in V$ represent an ambulance station. The service area for a time threshold t is defined as:

$$SA_t(s) = \{v \in V | d(s, v) \leq t\}$$

where $d(s, v)$ is the shortest path travel time from station s to node v . This formulation defines the set of all locations that can be reached within time t .

To evaluate overall system coverage, the union of service areas from all stations is computed:

$$SA_{total} = \bigcup_{k=1}^n SA_t(s_k)$$

This allows for the identification of both covered and uncovered regions within the study area.

In this study, multiple thresholds are considered (e.g., 5, 10, and 15 minutes) to represent different levels of accessibility. These thresholds are commonly used in emergency response planning to assess service efficiency.

Figure 3.4 illustrates the spatial distribution of ambulance service coverage based on predefined travel time thresholds. The diagram presents concentric service zones surrounding the ambulance station, representing areas that can be reached within 5 minutes, 10 minutes, and 15 minutes, respectively. These zones are generated using network-based service area analysis, where accessibility is determined by travel time along the road network rather than straight-line distance.

The innermost region indicates areas with the fastest response time, reflecting high accessibility and optimal emergency coverage. As the distance from the station increases, the response time also increases, as shown in the outer zones. The region beyond the 15-minute threshold represents areas with limited or delayed access to emergency services, highlighting potential gaps in coverage. This analysis is crucial for emergency planning, as it helps identify underserved regions and supports decision-making regarding the optimal placement of additional ambulance stations to improve overall response efficiency.

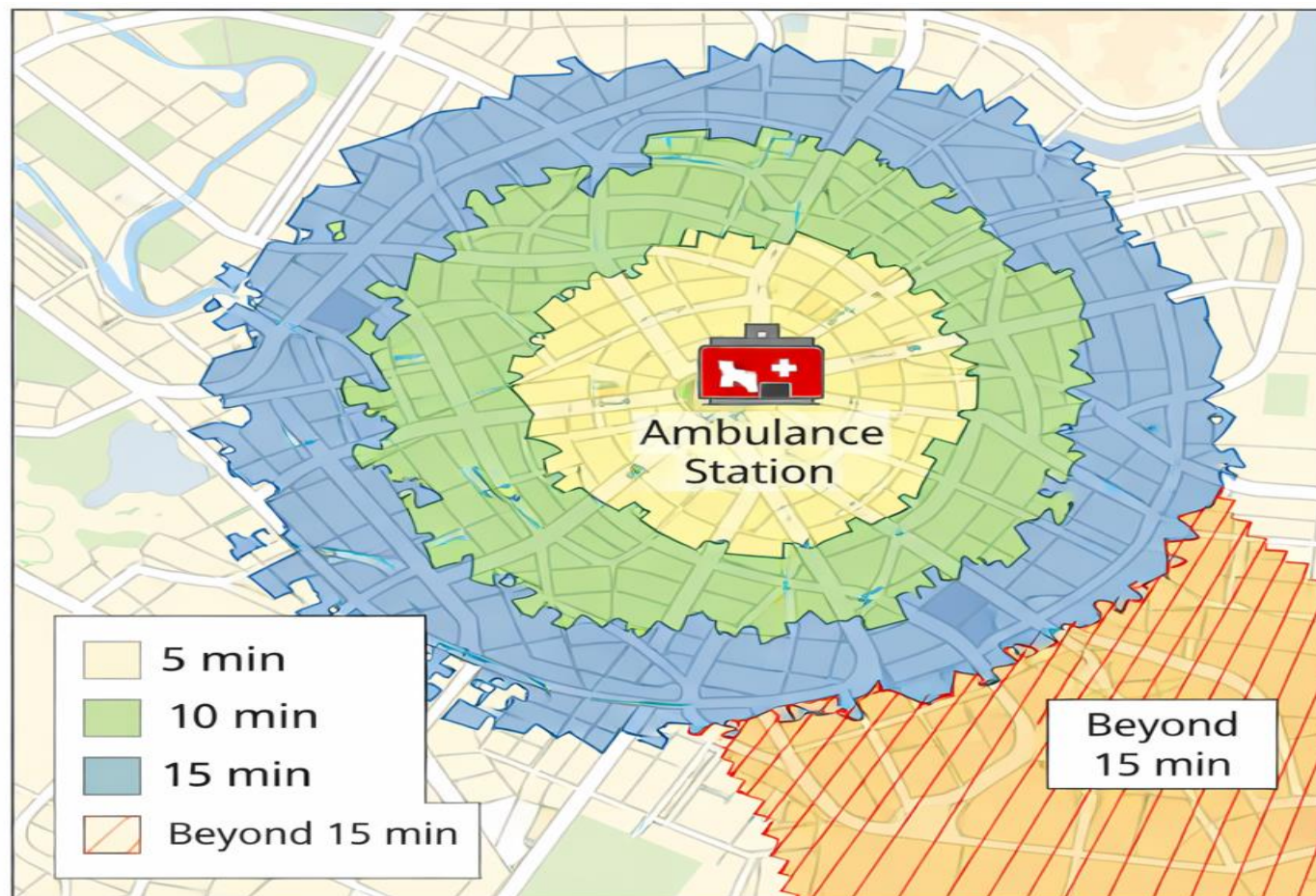


Figure 3.4: Service Area Coverage Map

3.8 System Architecture

The system architecture is designed to integrate spatial data processing with computational analysis in a structured and scalable manner. It follows a layered architecture consisting of three primary components: the data layer, the processing layer, and the output layer.

The data layer is responsible for storing all input datasets, including road networks, ambulance locations, and incident data. The processing layer implements network analysis operations such as shortest path computation, closest facility determination, and service area generation. The output layer presents the results in the form of maps, routes, and analytical metrics.

The interaction between these components can be expressed functionally as:

$$Output = f(Data, Algorithms)$$

where the function f represents the transformation of spatial data through network analysis algorithms.

Figure 3.1 illustrates the overall architecture of the proposed GIS-based ambulance routing and emergency response analysis system. The architecture is structured into three major layers: the data layer, the processing layer, and the output layer. The data layer comprises essential spatial datasets including the road network, ambulance station locations, and emergency incident points, which serve as the foundational inputs to the system.

The processing layer is responsible for transforming raw spatial data into actionable insights through network dataset construction and the application of network analysis algorithms. This layer integrates GIS tools to perform operations such as route optimization, closest facility analysis, and service area computation. The output layer presents the results in the form of optimized routes, service area maps, and response

time analytics, which are essential for decision-making in emergency management. The architecture demonstrates a clear flow of data from input through processing to output, ensuring efficiency, scalability, and effective emergency response planning.

3.9 Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) provides a logical representation of how data moves through the system and how different processes interact. It identifies the key inputs, processing stages, and outputs involved in the system.

The system receives spatial datasets as input, which are then processed through a series of transformations including data cleaning, network construction, and routing analysis. The outputs generated include optimized routes, response time metrics, and service area maps.

This transformation can be mathematically represented as:

$$O = f(P(f(D)))$$

where D is raw data, $f(D)$ represents preprocessing, P represents network analysis processes, and O is the final output.

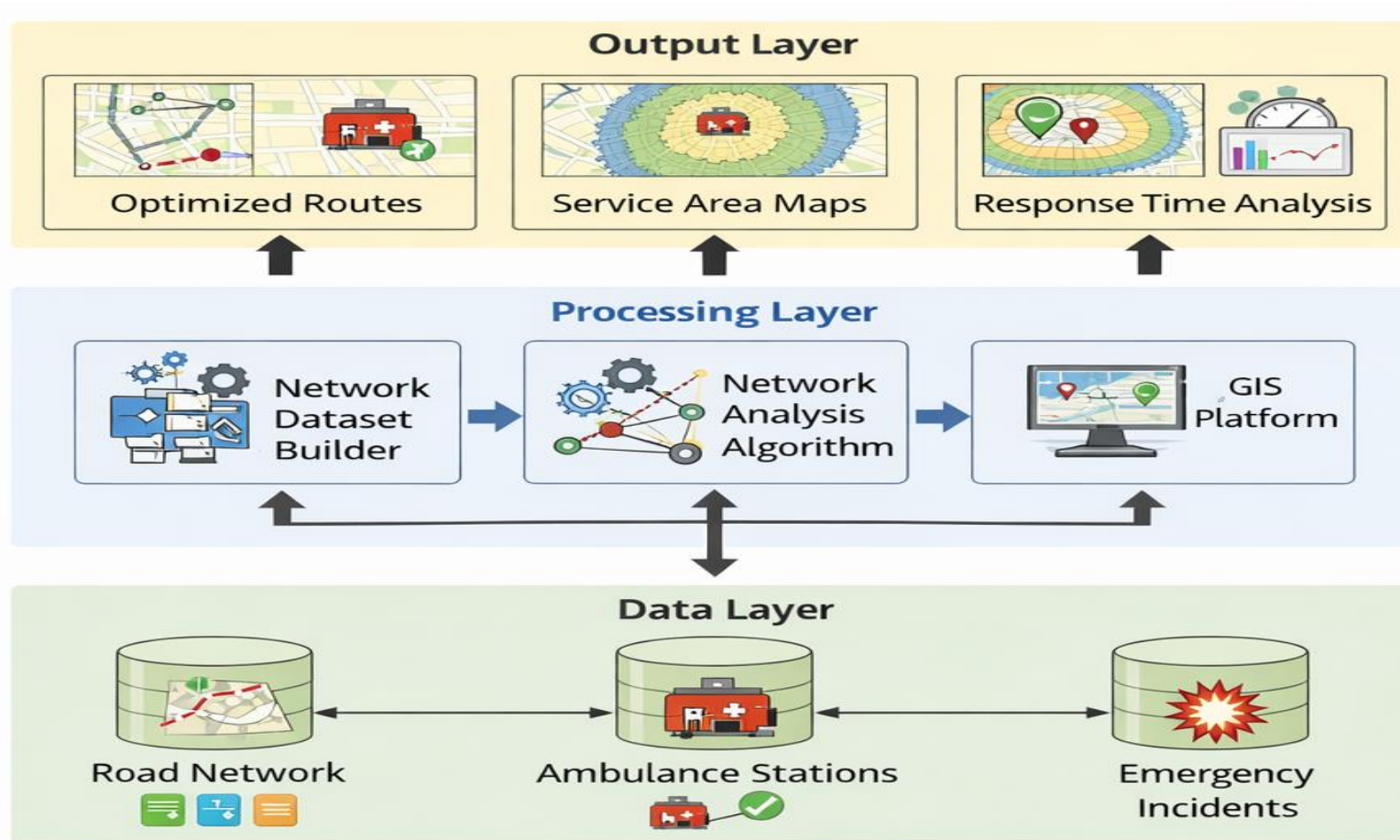


Figure 3.1: System Architecture of the Proposed Model

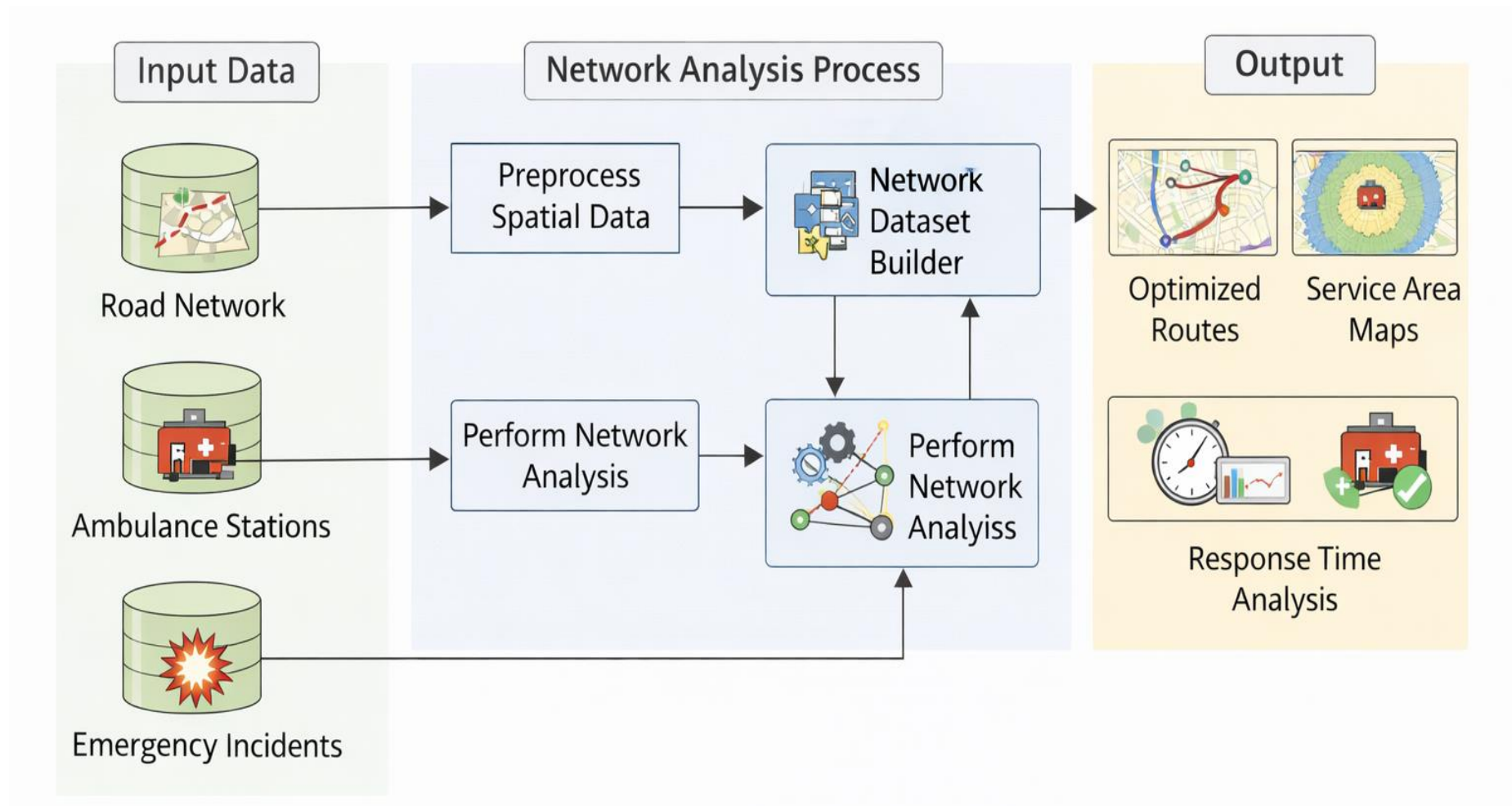


Figure 3.5: Data Flow Diagram of the System

Figure 3.5 presents the Data Flow Diagram (DFD) of the proposed GIS-based ambulance routing and emergency response system, illustrating how data moves through different stages of processing. The diagram begins with the input data layer, which consists of key spatial datasets including the road network, ambulance station locations, and emergency incident data. These inputs serve as the foundation for all subsequent analyses.

The data is then passed into the network analysis process, where preprocessing is first carried out to clean and structure the spatial data. This is followed by the construction of a network dataset, which enables the application of routing and accessibility algorithms. The system then performs network analysis operations such as shortest path computation, closest facility identification, and service area generation. Finally, the processed data is transformed into meaningful outputs, including optimized routes, service area maps, and response time analysis. The diagram clearly demonstrates the logical flow and transformation of data, ensuring that raw spatial inputs are effectively converted into actionable insights for improving emergency response efficiency.

3.10 System Flowchart

The system flowchart provides a step-by-step procedural representation of the workflow involved in the implementation of the proposed system. It begins with data acquisition and preprocessing, followed by network dataset construction and routing analysis. The process concludes with visualization and evaluation of results.

The workflow can be summarized as follows:

1. Input spatial data
2. Preprocess and clean data
3. Construct network dataset

4. Perform routing and accessibility analysis
5. Generate outputs and evaluate performance

Figure 3.6 illustrates the procedural workflow of the proposed GIS-based ambulance routing and emergency response system. The process begins with the input of spatial data, including road networks, ambulance station locations, and emergency incident points. This data undergoes preprocessing to ensure accuracy and consistency before being transformed into a network dataset suitable for analysis.

Subsequently, the system performs route optimization using shortest path algorithms to determine the most efficient paths for ambulance movement. This is followed by closest facility analysis to identify the nearest available ambulance station, and service area analysis to evaluate coverage and accessibility. Finally, the system generates outputs in the form of optimized routes, service area maps, and response time metrics. The flowchart provides a clear step-by-step representation of the system's operation, ensuring transparency, logical organization, and ease of implementation.

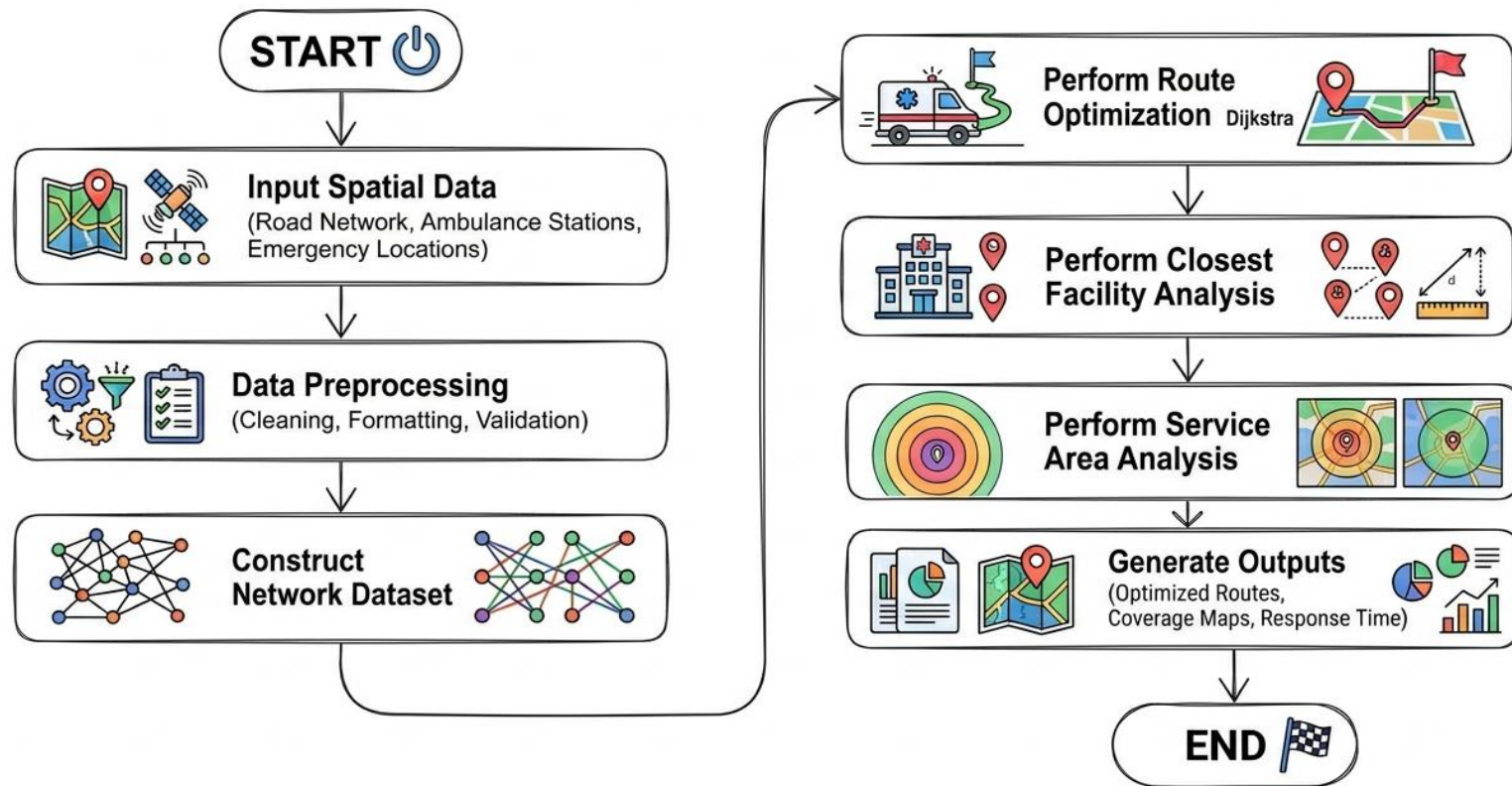


Figure 3.6: System Flowchart for proposed solution

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